Andrew Park

917.783.7056 | andrewpark.us@duke.edu | New York, NY

Education

Duke University, Trinity College of Arts & Sciences Durham, NC

*B.S. Mathematics, B.S. Computer Science; GPA **3.96/4*** *Expected May 2028*

- Won the Julia Dale Prize for first-year mathematical excellence, and Dean's List with Distinction.
- **Graduate Courses:** Mathematical Finance, Financial Derivatives, Real Analysis, Algebraic Structures, Applied Stochastic Processes (Fall 2026).
- **Undergraduate Courses:** Advanced Probability, Data Structures & Algorithms, Computer Architecture, Graphics Software Architecture.
- Graded graduate Mathematical Finance as an undergraduate TA, plus proof-based problem sets in linear algebra, multivariable calculus and differential equations.

Stuyvesant High School: ACT 36; National Merit Scholar New York, NY

Experience

Quantitative Research Intern, DataBallR May 2026 - Present

Daily RAPM, a public multi-league player valuation model at andrewjparkus.github.io

- Designed a Kalman rating model that grades **10,681** college, G League and NBA players on the same scale.
- Led the 2026-27 season projection. Player minutes, team depth charts, rotations and games missed each get their own model, rather than one black box that prints a number.
- Improved on the **Vegas closing line** for NBA game margins by blending our ratings into it. Over 12,034 games and 12 held-out seasons, the fit put about 30% of its weight on us.

PRUV Research Fellow, Duke Mathematics April 2025 - Present

Senior thesis under Professor J. Thomas Beale; manuscript in preparation *Durham, NC*

- Modeled atmospheric flow on rapidly spinning brown dwarfs, which bulge out of the spherical shape nearly every current paper in the field assumes.
- Directed a **4,500-line Julia** solver that puts the shallow-water equations on those non-spherical figures, using Beale's surface discretization and verified against exact solutions to second-order spatial accuracy. The rotational signal came out thousands of times stronger than the sphere predicts.

Quantitative Projects

Prediction-Market Trading & Election Forecasting June 2022 - Present

- Founded and led a seven-person team that built a U.S. election model on precinct returns back to 1912.
- Built a portfolio optimizer over it, sizing positions with **fractional Kelly** and hedging the national environment. Traded the book on Polymarket and Kalshi across presidential, Senate, House and gubernatorial races.

Duke Poker Club, Founder & President January 2025 - Present

- Re-launched the club to 100+ members, secured a collegiate Kalshi partnership and trained our intercollegiate tournament team. Interviewed by *The Wall Street Journal* on college poker and prediction markets.

Skills & Interests

Languages: Python, Julia, SQL, Java

Tools: NumPy, pandas, SciPy, scikit-learn, XGBoost, Arrow/Parquet, Git, LaTeX

Methods: probability, statistics, regression, stochastic processes, Monte Carlo, backtesting, portfolio optimization

Spoken: English and Korean (native), French (fluent)

Interests: classical piano; first violin, New York Youth Symphony (Grammy winner); nineteenth-century literature